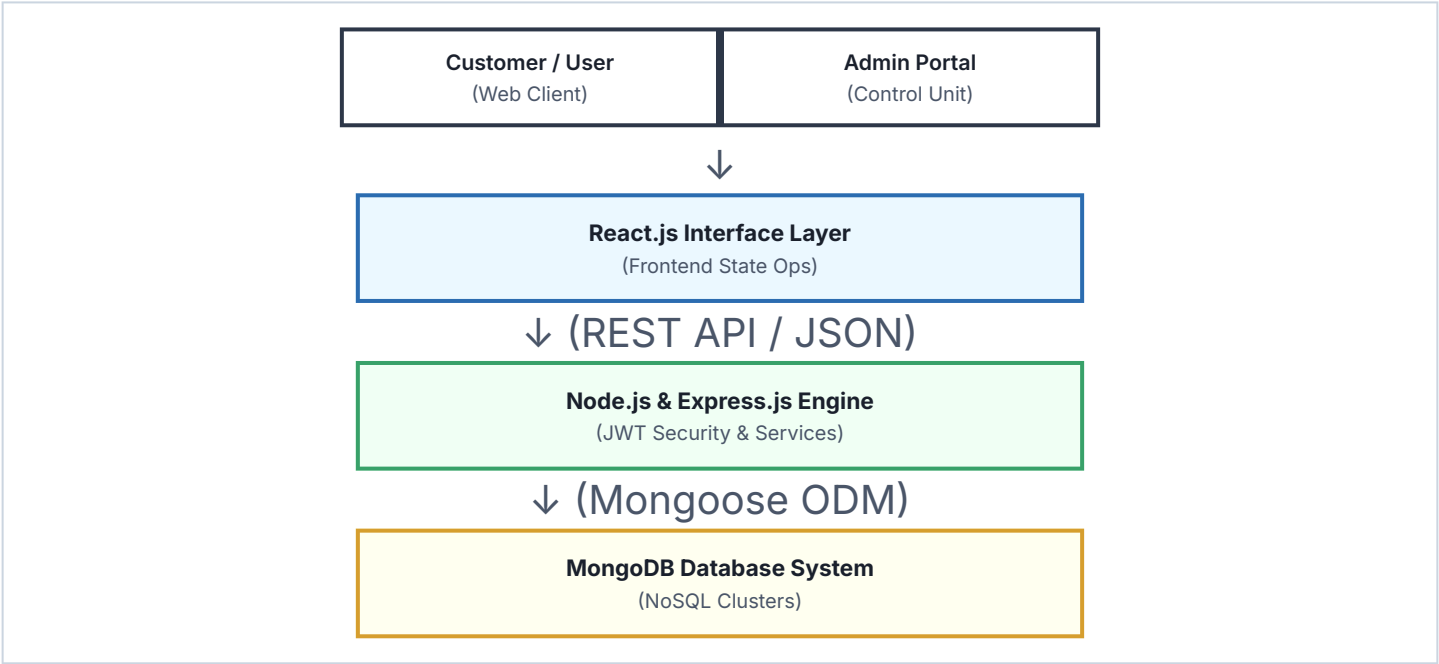


PROJECT DESIGN PHASE-II

Technology Stack & System Architecture

Date	18 June 2026	Project Name	ShopEz (MERN E-Commerce)
Team ID	--	Maximum Marks	4 Marks
Student Name	Turlapati Lahari (24eg109a54@anurag.edu.in)		

1. Technical Architecture (MERN Stack)



2. Components & Technology Stack

Component	Description	Technology
User Front-end	Single Page Hub for shopping and admin tasks.	React.js, Tailwind
Backend Logic	Governs API requests and business processes.	Node.js & Express
Core Data Hub	JSON-based document storage for products.	MongoDB Atlas
Auth Logic	Safe encryption and session verification.	JWT & Bcrypt.js
State Control	Manages user cart and session variables.	Context API
Hosting	Deployment and development environment.	Vercel / Heroku

3. Module Design & Data Architecture

Module	Functional Description	Key Database Fields
Auth Control	Manages signup, login, and token flows.	email, hash, role
Product Unit	Catalog browsing and inventory logic.	name, cost, category
Order Unit	Processing purchases and buyer logs.	userId, items[], total

4. API Interface Design (REST)

Endpoint	Method	Purpose
/api/users/login	POST	Validates login and grants JWT.
/api/products	GET	Pulls various catalog data points.
/api/orders/place	POST	Saves specific order records.
/api/admin/product	POST/PUT	Admin stock modification tools.

5. Design Considerations

Scalability: The MongoDB environment facilitates easy horizontal growth and schema adjustments as the inventory expands. **Security:** All data endpoints utilize JWT middleware for protection; user passwords remain hashed at all times. **Performance:** The Virtual DOM architecture in React enables high-speed rendering for real-time inventory filtering.